

Manual/Auto Station Installation and Operation

PM12-430

**Implementation
PM/APM/HPM Optional Devices**

Manual/Auto Station Installation and Operation

**PM12-430
Release 430
11/95**

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The following standard document conventions may be used in this document. Check Section 1 for conventions unique to this document.

Item	Convention	Meaning
<KEYCAPS>	Pointed brackets around a term	A key on a computer keyboard
[TARGET]	Square brackets around a term	For touch screen monitors, an area that can be touched to invoke a system action
entry data	Courier bold text	Data you enter into the system
screen/program data	Courier text	Screen or program data

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Manual/Auto Station Installation and Operation

Section 1—Introduction

1.1 About This Document

Basics

Purpose	Describes how to install a Manual/Auto Station, configure local setup functions, and perform operations.
Intended audience	Anyone installing and configuring a Manual/Auto Station.
For product release	TDC 3000 [®] R230 and later.

Product background

The Manual/Auto Station provides local panel display and manipulation capability for a single control loop residing in the Advanced Process Manager (APM) or Process Manager (PM).

M/A Stations are used as an alternate operator interface for process loops which may be located near the equipment being controlled.

The M/A Station is intended to provide backup operator interface for critical or selected control loops. The primary operator interface for loops with M/A Stations is the Universal Station(s).

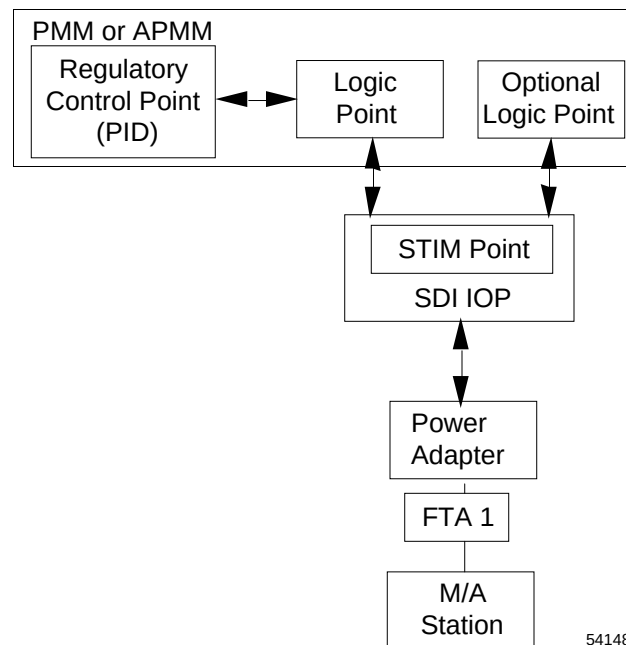
The M/A Station is designed for panel board mounting and can be located up to 1000 feet from the field termination cabinet (consult the factory for longer lengths).

1.2 M/A Station Functions

Configuration

When using an M/A Station, the Serial Device Interface (SDI) I/O Processor (IOP) of the PM or APM must be configured as a Smart Transmitter Interface Module (STIM) board type. An STIM point and one or two logic points must also be built for each M/A Station (see the following figure). Local configuration must also be performed at the M/A Station.

Signal Flow



Single control loop functions

The M/A Station provides the following functions for a single control loop:

- Three vertical bar graphs and two alphanumeric digital displays present current loop process signals to the operator (PV, setpoint, and output).
- Acts as a tactile feedback keyboard for changing the setpoint, output, or mode.
- Allows manual control of the setpoint or output signal and the loop mode (manual or automatic).
- Gives a visual indication of the loop mode and alarms.
- Allows configuration of alphanumeric descriptors, which appear in the M/A Station's digital displays.

Source of data

The M/A Station operates as a slave device to the APM/PM control loop. Signals to and from the process are wired to the APM/PM, not to the M/A Station. The process and status signals are transmitted serially from the APM/PM to the M/A Station(s) through an EIA-485 communication link. (Note: RS-485 is now officially ANSI/EIA-485.)

The M/A station receives all operational data such as PV, setpoint, output, and alarms from the Advanced Process Manager or Process Manager. The operator can only change the setpoint, mode, and output.

1.3 1.2 M/A Station Architecture

Security

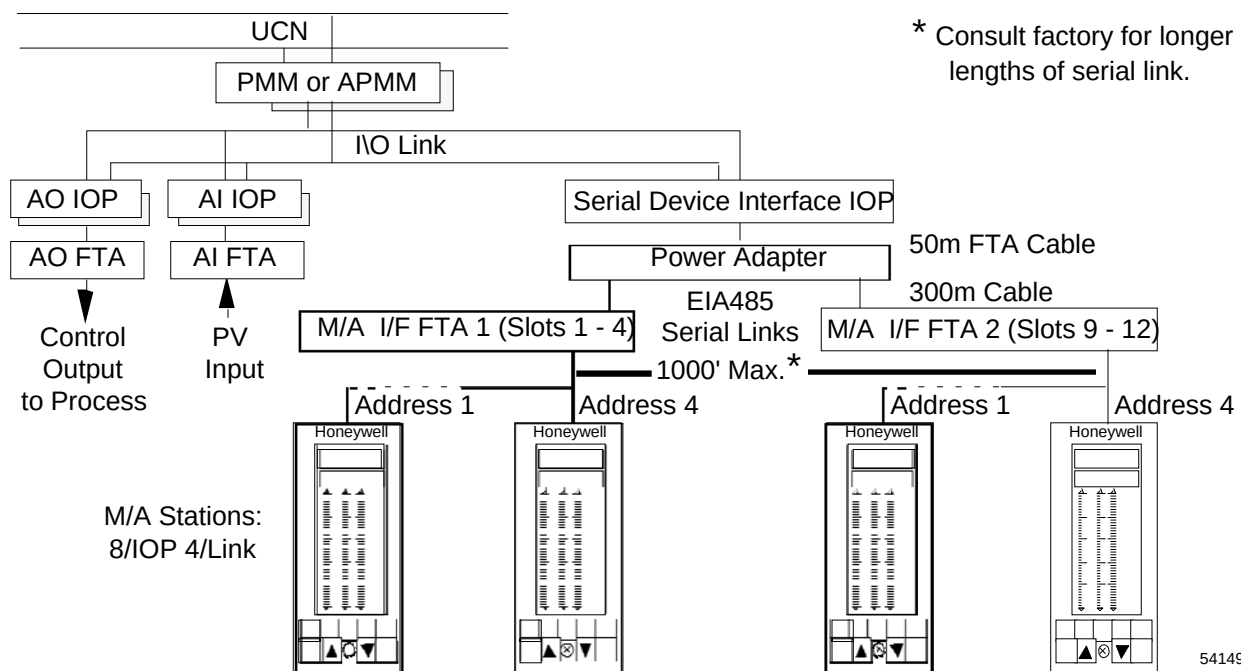
For increased process security and reliability of the regulatory control PID loop, the analog output to the final control element can be made redundant. If the loop uses a high level analog input, it can also be made redundant, if desired.

Addressing

As illustrated in the following figure, each M/A station Field Termination Assembly (FTA) can communicate with up to four M/A station units through an EIA-485 link. Since each SDI IOP can have up to two FTAs, a total of eight M/A stations can be connected to each SDI IOP.

Although not all eight slots are used, the slot numbers (1 through 8) in the FTA correspond to the addresses of the appropriate M/A station units. An STIM point built against FTA slot 3 communicates with an M/A station configured locally as address 3 on the EIA-485 link.

M/A station system architecture



1.4 References

Related information Documents containing related Manual/Auto station information are described in the following table.

Table 1-1 Related Topics Listing

These documents...	Contains these M/A station topics
PM and APM Control Functions and Algorithms documents	Descriptions of the communications between the APM/PM and the Manual/Auto station Lists of point parameters used to transfer data to/from the Manual/Auto station
PM and APM Implementation Guidelines documents	Descriptions of the PM/APM logic point(s) and STIM point configuration required to support the Manual/Auto station
PM and APM Parameter References Dictionaries	Definitions of STIM point parameters that apply to the Manual/Auto station
PM and APM Site Planning documents	Physical layout diagram of SDI FTA Interface cable pin assignments
PM and APM Installation documents	Wiring and cabling diagrams
PM and APM Service documents	SDI IOP Status Display example Part numbers

Document access and listing Throughout this document, references to other documents are made by title only.

- To access documents provided electronically, select the document title in the main topics list (Select Main from the Topics button).
- To access anything in the paper documentation, find the title in the following document listing, and use the binder number provided there.

Table 1-2 Document Listing

Document Title	Binder Number	Binder Name
Process Manager Installation	2040-1	Implementation/Process Manager - 1
Process Manager Control Functions and Algorithms	2040-2	Implementation/Process Manager - 2
Process Manager Service	2061	PM/APM Service
Process Manager Implementation Guidelines	2040-1	Implementation/Process Manager - 1
Process Manager Parameter Reference Dictionary	2040-3	Implementation/Process Manager - 3
PM and APM Planning	2020-1	System Site Planning - 1

Introduction

1.4 References

Table 1-2 Document Listing (continued)

Document Title	Binder Number	Binder Name
Advanced Process Manager Installation	2042-1	Implementation/Advanced Process Manager - 1
Advanced Process Manager Control Functions and Algorithms	2042-2	Implementation/Advanced Process Manager - 2
Advanced Process Manager Service	2061	PM/APM Service
Advanced Process Manager Implementation Guidelines	2042-1	Implementation/Advanced Process Manager - 1
Advanced Process Manager Parameter Reference Dictionary	2042-3	Implementation/Advanced Process Manager - 3

Section 2—M/A Station Installation

2.1 Before Installing the M/A Station

Overview	There are several steps you must take before actually installing the M/A Station. After checking your equipment for any shipping damage, the Power Select Kit containing the proper fuses, and power select jumpers must be installed. Once this kit is installed, the M/A Station can be mounted.
Equipment inspection	Remove the M/A station from its shipping carton and inspect the unit for any obvious shipping damage. Report any damage that occurred during transit to the carrier. Also check that the carton includes a bag of M/A Station mounting hardware.
Operating conditions	Review the following data before installing the M/A station. • Temperature - 58 to 131°F (15 to 55°C) • Vibration - 0.2G rated, 0.5G shipping • Power - – Voltage: 24 Vdc, 120 Vac or 240 Vac – Frequency: 50 or 60 Hz $\pm$2 Hz – Consumption: 13 Va
Power options	The M/A Station has a universal power option so that you can use either 120 Vac or 240 Vac, or 24 Vdc. Consult the low volume operation (LVO) factory for power distribution if +24 Vdc operation is required.

2.2 Installing the Power Select Kit

Kit contents The Power Select Kit supplied with the M/A Station contains these parts:

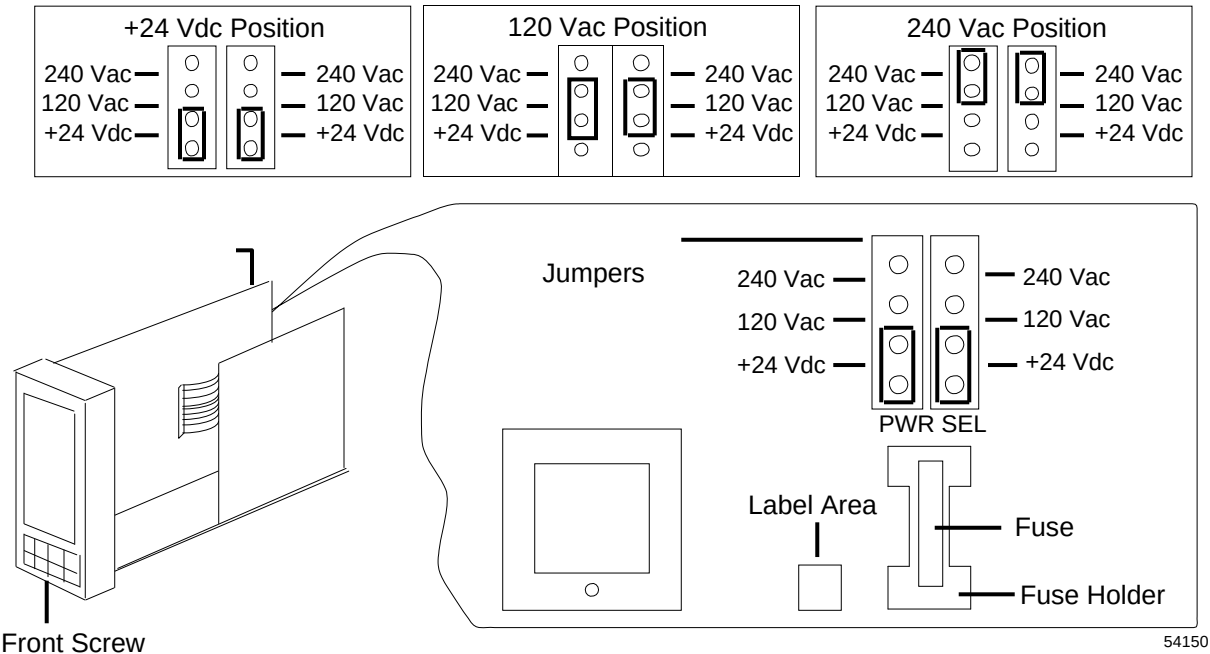
- 1 Fuse, Slow-blow, 0.5 Amp
- 1 Fuse, Slow-blow, 2.0 Amp
- 2 Power Select jumpers

Procedure Use the following procedure to install the Power Select Kit (30755498-003).

Table 2-1 Power Kit Installation Procedure

Step	Action
1	Loosen the front screw and pull out the chassis.
2	Lay the chassis on a nonconductive surface.
3	Select the proper fuse for the voltage being used: • +24 Vdc - 2.0 amp fuse • 120 Vac and 240 Vac - 0.5 amp fuse
3	Push the fuse into the fuse holder clip on the controller board (Reference: Power select jumpers and fuse diagram, page 9).
4	Peel the backing off the fuse label supplied in the kit, and place the proper value (1/2 or 2) into the label area next to the fuse clip (Reference: Power select jumpers and fuse diagram, page 9).
5	Install the 2 power select jumpers into the correct position on the controller board for the voltage used (Reference: Power select jumpers and fuse diagram, page 9).
6	Replace M/A Station chassis into case and tighten the screw.

Power select
jumpers and fuse
diagram

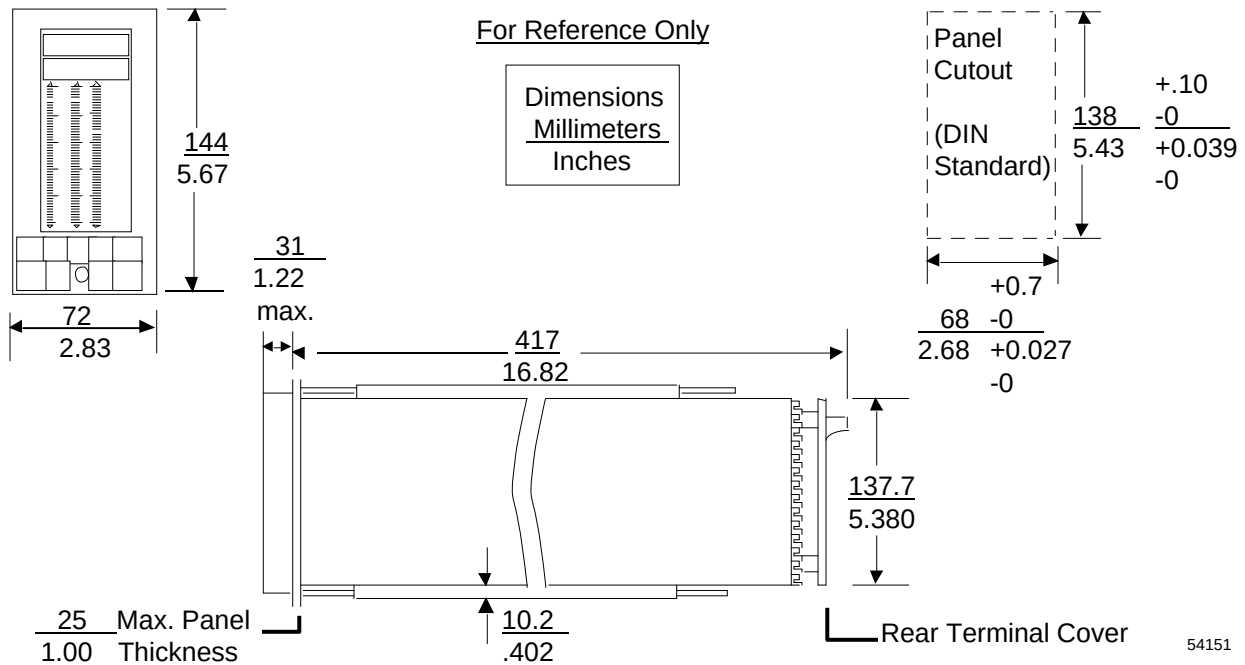


2.3 Mounting the M/A Station

Physical considerations

The M/A Station mounting kit (30755471-001) allows you to mount the station either on a vertical or tilted panel. There must be adequate space at the back of the panel to initially install the M/A Station and to service it in the future. The overall dimensions and panel cutout requirements for mounting the M/A Station are shown in the following figure.

M/A station dimensions



Mounting Procedure

Use the following procedure to install the M/A Station mounting kit.

Table 2-2 Mounting Procedure

Step	Action
1	Loosen the screw on the front of the M/A Station and pull the chassis out from the case.
2	Orient the case properly and slide it through the panel hole from the front.
3	Take the screws from the mounting kit and install them into the threaded holes of the clips.
4	Insert the prongs of the clips into the 2 holes in the top and bottom of the case.
5	Tighten both screws to secure the case against the panel.
6	Carefully slide the chassis assembly into the case, press to close, and tighten the front screw.

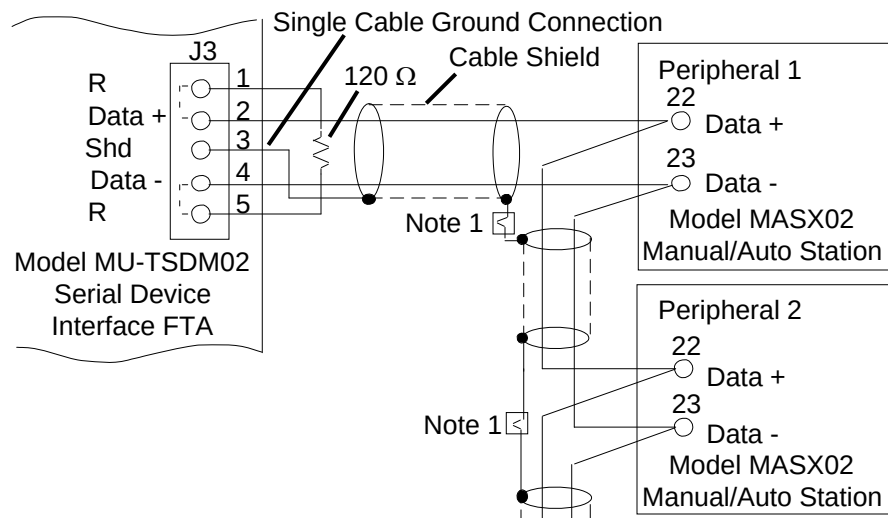
Wiring

Use the guidelines in the following table for wiring shields, wiring power, and routing cables. See *M/A station wiring to FTA—part 1*, page 11/ *M/A station wiring to FTA—part 2*, page 12 and *M/A station connection (rear view)*, page 12 for wiring diagrams.

Table 2-3 Wiring Guidelines

Shield Wires	- Connect the shield wires together with a crimp connector (Honeywell part number 30755381-001) - Ground the cable shield at only one location by connecting the shield wire to J3 on the FTA (pin 3 as shown in <i>M/A station wiring to FTA—part 1</i>, page 11/ <i>M/A station wiring to FTA—part 2</i>, page 12). The last unit on the link is NOT grounded. - Insulate the exposed ends of the shield wire.
Cable Routing	- Do not route the cables in the same conduit or tray as AC power. - The cable length from the M/A station to the APM/PM can be a maximum of 1000 feet.
Power	- Ground each M/A station separately to AC safety ground using 12 AWG wire. - If the source of power has a grounded neutral, connect L2 directly to AC Neutral (Reference: <i>M/A station connection (rear view)</i>, page 12). - Consult factory low volume operations (LVO) for power distribution if +24 V operation is required.

M/A station wiring to FTA—part 1



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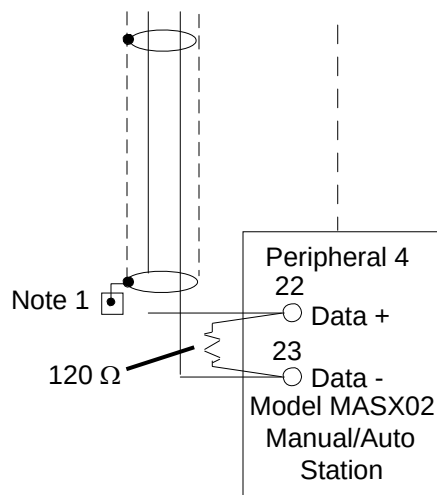
M/A Station Installation

2.3 Mounting the M/A Station

M/A station wiring to FTA—part 2

Notes

1. Connect the shield wires together with a crimp connector, Honeywell part number 30755381-001. Ground the cable shield at only one location by connecting the shield wire to J3, pin 3 on the FTA as shown. Insulate the exposed ends of the shield wire.
2. Do not route the cables in the same conduit or tray as AC power.
3. J3, pin 1 is connected to pin 2, and J3, pin 4 is connected to pin 5 internally on the FTA.
4. Up to four model MU-MASX02 Manual/Auto Stations can be connected to the EIA-422/485 interface.



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M/A station connection (rear view)

Model MU-MASX02 Manual/Auto Station			
○	1	14	○
○	2	15	○
○	3	16	○
○	4	17	○
○	5	18	○
○	6	19	○
○	7	20	○
○	8	21	○
○	9	22	○
○	10	23	○
○	11	⊥	○
○	12	L1	○
○	13	L2	○

Notes:

1. Ground each device separately to AC Safety Ground using 12 AWG wire.
 2. If the source of power has a grounded neutral, connect L2 directly to AC Neutral.
 3. Consult the Honeywell low volume operations factory for power distribution if +24 volt operation is required.
- Data +
Data -
AC Safety Ground (Note 1)
AC Hot or +DC (Note 3)
AC Neutral or -DC (Notes 2 and 3)

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Bar graphs

The M/A station has three bar graphs that indicate current values as defined in the following table. These values are updated each time the link is scanned by the APM/PM (The link operates asynchronously rather than on a fixed scan period.).

Table 3-1 Bar Graph Indicators

Value	Location	Indication
Setpoint	Left side	Setpoint in percent. Operates as configured: • column of lighted bars, or • single tick mark. (Note: If setpoint is not used, this bar graph remains off.)
Process Variable	Center	Process Variable in percent. Column of lighted bars.
Output	Right side	Output in percent. Column of lighted bars.

Upper display

During normal operation, the upper digital display;= presents the current PV value and an optional 3-character identifier referred to as the PV tag.

Pressing the [SET UP] key places the M/A station in a mode that allows local configuration setup. In the setup mode, pressing the [FUNC] key displays local configuration data in the upper display.

The upper display is described in the following table.

Table 3-2 Upper Display Indicators

Indication	Format	Range	Example
PV Identifier and PV value	XXX YYYY X = 3 alphanumeric characters Y = PV value	-999.9 -99.99 -9.999 0.0 9.999 999.9 9999	FIC 300.0
Configuration Selections	values or prompts	(see 4-1 Configuration Procedure, page 19)	(see 4-1 Configuration Procedure, page 19).

Lower display

The lower alphanumeric digital display presents various types of information:

- **Configuration Functions**
If the [SETUP] key is pressed, the lower display shows configuration functions (see 4.1 Configuring Set Up Functions, page 19).
- **Error Messages**
In the event of a special condition, error messages will flash between the normal lower display and the error message (see 3.2 Error Messages, page 17).

- **Function Key Test**
If the [SETUP] and [FUNC] keys are pressed simultaneously, the keynames are displayed in the lower display after a display test has completed.
- **Normal Operating Information**
With each press of the [LOWR DISP] key, one of the operating parameters described in the following table, is displayed in the lower display.

In the following table, the value appearing in the Valve Position location can be any value the user decides to send from the APM/PM. For the value to represent the actual valve position, that variable would have to be available to the APM/PM using an input point, then an APM/PM logic point can be configured to send the value down to the M/A station.

Table 3-3 Lower Display Indicators

Indication	Format	Range	Example/Comment
Setpoint	SP_YYYY	same as PV	SP_300.0
Deviation	DEV_YYYY		DEV_
Output	OP_YYYY	-5 to 105.0	OP_50.0 (fixed at one decimal place)
Valve Position	VP_YYYY		VP_
Loop Identifier	up to 8 characters	alphanumeric and special characters	FIC 5000

Alarm annunciators

Two alarm status annunciators appear on the operator panel as either a lighted “1” or “2,” or both. The annunciators light when an alarm status is received from the APM/PM. A logic point in the APM/PM determines the conditions under which the lamps are lit.

Mode annunciators

Two mode annunciators appear on the operator panel as described in the following table:

Table 3-4 Mode Indicators

Indication	Description
MAN	Lit when the M/A station mode is Manual.
AUTO	Lit when the M/A station mode is Automatic
Flashing MAN or AUTO	Flashing occurs after the M/A station operator requests a mode change. Flashing continues for a minimum of 5 seconds until the new mode status is received from the APM/PM.

Function keys

The following table describes the operator panel function keys.

Table 3-5 Function Keys

Key	Description
[SET UP]	Puts the M/A Station in the set up mode. Once this key is pressed, the [FUNC] key can be used to sequentially display the setup functions. The M/A Station continues to communicate with the APM/PM during the set up mode.
[FUNC]	After pressing the [SET UP] key, the [FUNC] key is used to select individual parameters for each set up function. Each time the key is consecutively pressed, an additional function and description is shown in the displays. The upper display shows alphanumeric values, while the lower display shows parameters names.
[LOWR DISP]	Displays the operating information needed to monitor your M/A Station. Each time the key is pressed, the next parameter is shown in the lower display (SP, deviation, output, valve position, loop identifier).
[MAN/AUTO]	Toggles the M/A Station between the auto and manual modes. The annunciators indicate the mode selection.
 	Used to change the value for the operating parameter that is shown in the lower display (SP, OP, or VP). These keys also used to change the configuration selection displayed the upper display.
Note: When you press one of these keys to change a number, the least significant digit increments or decrements. To accelerate the increment or decrement, tap (momentarily press) the opposite key (if incrementing, tap the decrement key. If decrementing, tap the increment key). Tapping the opposite key increases the value of the display by 10 each time. You must hold one key down while the other is tapped.	
[SET UP] plus [FUNC]	When these two keys are pressed at the same time, a display test occurs in which display segments (even those not normally used by the M/A Station) are on. The display test is then followed by a key test. When you press a particular key, the key name is shown in the lower display. Note: The M/A Station is capable of communicating with the APM/PM during the display and key tests.
[RSP/CAS], [SP1/SP2], [ALM/ACK]	These keys are not used in this application. Pressing these keys results in a KEY ERROR message in the lower display.

3.2 Error Messages

Display In the event of an error condition, messages flash in the lower display. These messages normally flash at a 70% duty cycle (7 seconds on, then 3 seconds off).

Definition There are two possible error messages that can display:

- **OVERRIDE**
The M/A station supports an override flag which denies permission for an operator to change the Mode, Setpoint, or Output from the M/A station.
When the M/A station is placed in override from the APM/PM, the lower display flashes the **OVERRIDE** message.
If a change is attempted while override is on, the **KEY ERROR** message appears.
The override flag can be set by the operator at a Universal Station using a flag point configured for that purpose, or a program, or logic point can set the override flag based on certain conditions.
- **COM_ERROR**
This message appears whenever the M/A station detects an error in its own hardware, or has not received a valid communication over the EIA-485 link in the past 5 seconds. During the **COM_ERROR** period, all data and statuses are held at the last value.
For example, if the STIM interface point in the APM/PM is inactive, a **COM_ERROR** occurs.

Section 4—Configuration and Operation

4.1 Configuring Set Up Functions

Function list Performing set up functions allows you to specifically configure your M/A Station for your particular needs. The following functions can be configured on the M/A Station:

- Setpoint Display (Bar graph or tick mark)
- 8-character Identification Tag
- 3-character Process Variable Tag
- Decimal Place
- Station Address

Keystroke overview

Use the following keystrokes to configure set up functions.

1. Press [SETUP] to enter set up mode.
2. Press [FUNC] until desired function is displayed in lower display.
3. Press [▼] or [▲] to change selection displayed in upper display.
4. Repeat steps 2. and 3. until all functions display the desired selections.
5. Press [LOWR DISP] to exit the set up mode at any time.

Timeout occurs after one minute if no key is pressed. The M/A station then reverts back to the mode and display previously being used.

Configuration procedure

The following table describes the M/A station configuration procedure.

Table 4-1 Configuration Procedure

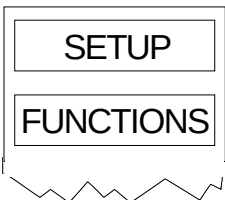
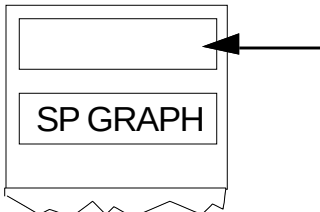
Press	Result: Upper/Lower displays	Press [▼] [▲] to display selection	Remarks
[SET UP]		not applicable	LOCAL SET UP MODE. Press [FUNC] key to display desired parameter.
[FUNC]		Selections: Full One Bar	SETPOINT BAR GRAPH — Makes the bar graph a single tick mark or a column of bars: FULL - bars are lit up to the value ONE BAR - tick mark is indicated at value

Table 4-1 Configuration Procedure (continued)

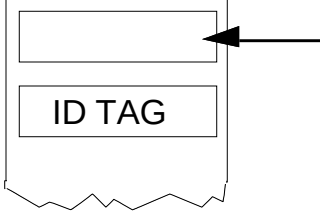
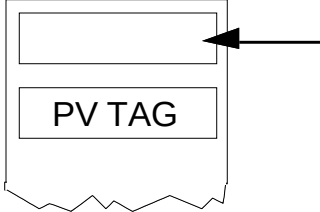
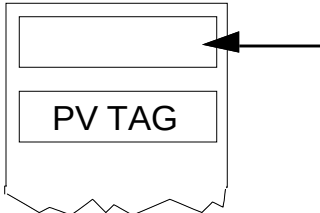
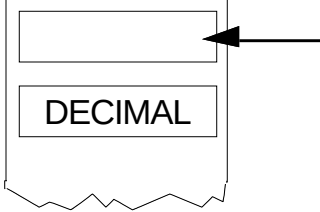
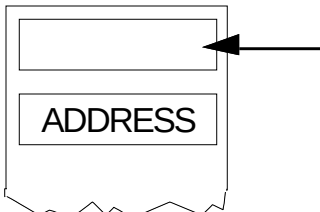
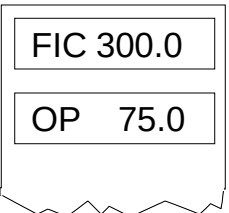
Press	Result: Upper/Lower displays	Press [▼] [▲] to display selection	Remarks
[FUNC]		Selections: Enable Disable	IDENTIFICATION TAG — Enables entry of 8-character loop identifier: ENABLE - enables ID tag (see following) DISABLE - disables ID tag (see following)
[FUNC]		Selections: Letters: A to Z Numbers: 0 to 9 Others: >, <, #, /, □, (blank), -, ., °, =, ?, _ , +, *	IDENTIFICATION TAG — Enters an 8-character identifier that is displayed during normal operation using [LOWR DISP].
NOTES: - The currently active character position blinks. Every three seconds, the M/A station moves on to the next character unless changes are made. - The STIM point parameter STITAG is also used to define the Tag ID string. This parameter is automatically downloaded to the M/A station from the APM/PM when the STIM point is made active.			
[FUNC]		Selections: Enable Disable	PV TAG — Enables entry of a three-character PV identifier: ENABLE - enables PV tag (see following) DISABLE - disables PV tag (see following)
[FUNC]		Selections: Letters: A to Z Numbers: 0 to 9 Others: >, <, #, /, □, (blank), -, ., °, =, ?, _ , +, *	PV TAG — Enters a 3-character identifier that is always displayed during normal operation in the upper display beside the PV value. NOTES: The currently active character position blinks. Every three seconds the M/A station moves on to the next character unless changes are made.

Table 4-1 Configuration Procedure (continued)

Press	Result: Upper/Lower displays	Press [▼] [▲] to display selection	Remarks
[FUNC]		Selections: XXXX XXX.X XX.XX X.XXX	DECIMAL PLACE — Determines where the decimal point appears in the display of setpoint and process variable: XXXX - no decimal places XXX.X - one decimal place XX.XX - two decimal places X.XXX - three decimal places
[FUNC]		Selections: 1 through 4	STATION ADDRESS — Selects an address for this specific M/A station on the EIA-485 link. The address must be unique to other M/A stations on the link. Example: An APM/PM STIM point assigned to slot 3 communicates with the M/A station configured locally as address 3.
[LOWR DISP]		not applicable	Exits the set up mode and returns to previous display.

4.2 Changing the Modes

Procedure

You can change the mode of the M/A Station from manual to auto and vice versa by pressing the [MAN/AUTO] key. This key acts as a toggle from one mode to the other.

The following procedure is used to change the mode of the M/A Station.

Table 4-2 Mode Change Procedure

Step	Action	Result
1	Press the [MAN/AUTO] key.	The mode annunciator flashes for two seconds, or until the APM/PM confirms the mode. Note: During this period when the mode annunciator is either flashing or solid, the M/A Station ignores output updates from the APM/PM.
2	Has APM/PM confirmed the mode change? If yes, The annunciator stops flashing and remains on. The lower display changes to show the output value. (if MAN mode) or the setpoint value (if AUTO mode) If no, The 15 second time out period expires, and the previous mode annunciator stops flashing and turns off. Then the annunciator turns on.	Note: If the mode doesn't change, it may be because External Mode switching has not been enabled for the regulatory control point.

Restrictions

The mode can be changed whenever the M/A Station does not have the OVERRIDE error message displayed.

If you try to change to the manual mode when the M/A Station is in the OVERRIDE error condition, the message KEY ERROR is shown in the lower display.

4.3 Changing the Setpoint and Output

Restrictions The setpoint value can be changed in manual *or* automatic mode, *unless* PV Tracking is configured on the APM/PM regulatory control point, in which case, setpoint changes are not allowed in manual mode.

The output value can be changed *only* when the M/A station is in Manual mode; however, if the APM/PM regulatory control point is in initialization mode, the logic point that maps M/A station information to the regulatory control point is NOT allowed to alter any control point parameters.

The setpoint and output can be changed whenever the M/A station is NOT in OVERRIDE. If you try to change the setpoint when the M/A station is in OVERRIDE, the message KEY ERROR is shown in the lower display.

Setpoint procedure The following table describes the procedure to change the setpoint.

Table 4-3 Setpoint Change Procedure

Step	Action	Result
1	Press the [LOWR DISP] key until Setpoint (SP) value is shown in the lower display.	Current setpoint is displayed.
2	Press the [▼] or [▲] key until Setpoint is at desired value. Note: The range of the setpoint is limited only by the maximum allowable display range of +9999 to -999.0.	Setpoint is changed.

Output procedure The following table describes the procedure to change the output.

Table 4-4 Output Change Procedure

Step	Action	Result
1	If mode annunciator is not MAN, press [MAN/AUTO] key to place loop in Manual mode.	Mode annunciator is MAN.
2	Press the [LOWR DISP] key until Output (OP) value is shown in the lower display.	Current output is displayed.
3	Press the [▼] or [▲] key until Outputs at desired value. Note: The range of the output is only limited by the maximum allowable range of +105.0 to -5.0%.	Output is changed.

Status message to APM/PM When a change is made to either the setpoint or output value, the M/A station sends a status to the APM/PM during a normal transaction response, indicating the parameter being changed and the value. The status change indicator remains active for 15 seconds following the final key press.

Configuration and Operation

4.3 Changing the Setpoint and Output

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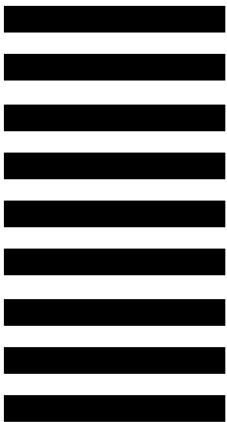
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